

Mohamed Mohamed

Software Engineer

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EDUCATION

George Mason University – Fairfax, VA **May 2026**

- Bachelor of Science in Computer Science, GPA: 3.33 | Dean's List: Spring 2025, Summer 2025
- Selected Coursework: Operating Systems, Analysis of Algorithms, Secure Systems & Programming, Database Concepts, Web App Development

Northern Virginia Community College – Annandale, VA **June 2023**

- Associate of Science in Computer Science, GPA: 3.71 | Magna Cum Laude, Honors Program, Phi Theta Kappa

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, SQL, HTML/CSS

Frameworks and Tools: Flask, SQLAlchemy, Jinja2, Bootstrap, Git/GitHub, Linux, Docker, OpenSSL, Wireshark

Technical Areas: REST APIs, relational databases, authentication, systems programming, TLS/PKI

PROJECT EXPERIENCE

OutfitPartPicker — *Python, Flask, SQLAlchemy, REST APIs, HTML/CSS, Bootstrap*

- Built a full-stack outfit discovery application with user registration, login, API-powered clothing search, and shareable outfit posts.
- Designed relational database models for users, outfits, clothing pieces, likes, comments, and feedback, including many-to-many relationships.
- Implemented multi-item outfit creation workflows and added caching for external product searches to reduce redundant API calls.

OS/161 Process Management & System Calls — *C, OS/161, System/161, GDB, Linux*

- Implemented getpid, getppid, fork, execv, waitpid, and _exit in an instructional operating system kernel.
- Managed parent-child process relationships, address-space copying, user argument transfer, process termination, and exit-status handling.
- Debugged behavior with GDB and automated test scripts, validating syscall correctness across normal and edge-case executions.

Code Generation Model Evaluation Pipeline — *Python, Hugging Face Transformers, Colab, Pandas*

- Built a unified evaluation pipeline to compare three open-source code generation models: CodeGen-350M, DeepSeek-Coder-1.3B, and StarCoder2-3B.
- Designed HumanEval-style Python tasks with function specifications and unit tests, then executed model-generated code in an isolated environment to measure pass@1 accuracy.
- Developed an interactive demo tool that accepts custom coding prompts and tests each model's generated solution automatically.
- Analyzed model reliability and failure modes, finding benchmark accuracies of 25%, 100%, and 50% across the evaluated models.

TLS/PKI Security Labs — *OpenSSL, Python, Docker, Wireshark, TLS/SSL*

- Built and analyzed secure client-server communication workflows using certificate authorities, CSRs, signed certificates, and HTTPS configuration.
- Inspected TLS handshake behavior, certificate validation, and trust-chain failures using Wireshark and OpenSSL tooling.
- Studied man-in-the-middle risk scenarios in controlled lab environments by comparing valid and compromised trust assumptions.